

FeatInv: Spatially resolved mapping from feature space to input space using conditional diffusion models

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FeatInv: Spatially resolved mapping from feature space to input space using conditional diffusion models

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Abstract: Internal representations are crucial for understanding deep neural networks, such as their properties and reasoning patterns, but remain difficult to interpret. While mapping from feature space to input space aids in interpreting the former, existing approaches often rely on crude approximations. We propose using a conditional diffusion model - a pretrained high-fidelity diffusion model conditioned on spatially resolved feature maps - to learn such a mapping in a probabilistic manner. We demonstrate the feasibility of this approach across various pretrained image classifiers from CNNs to ViTs, showing excellent reconstruction capabilities. Through qualitative comparisons and robustness analysis, we validate our method and showcase possible applications, such as the visualization of concept steering in input space or investigations of the composite nature of the feature space. This approach has broad potential for improving feature space understanding in computer vision models.

Submission Length: Regular submission (no more than 12 pages of main content)

Code: <https://github.com/AI4HealthUOU/FeatInv>

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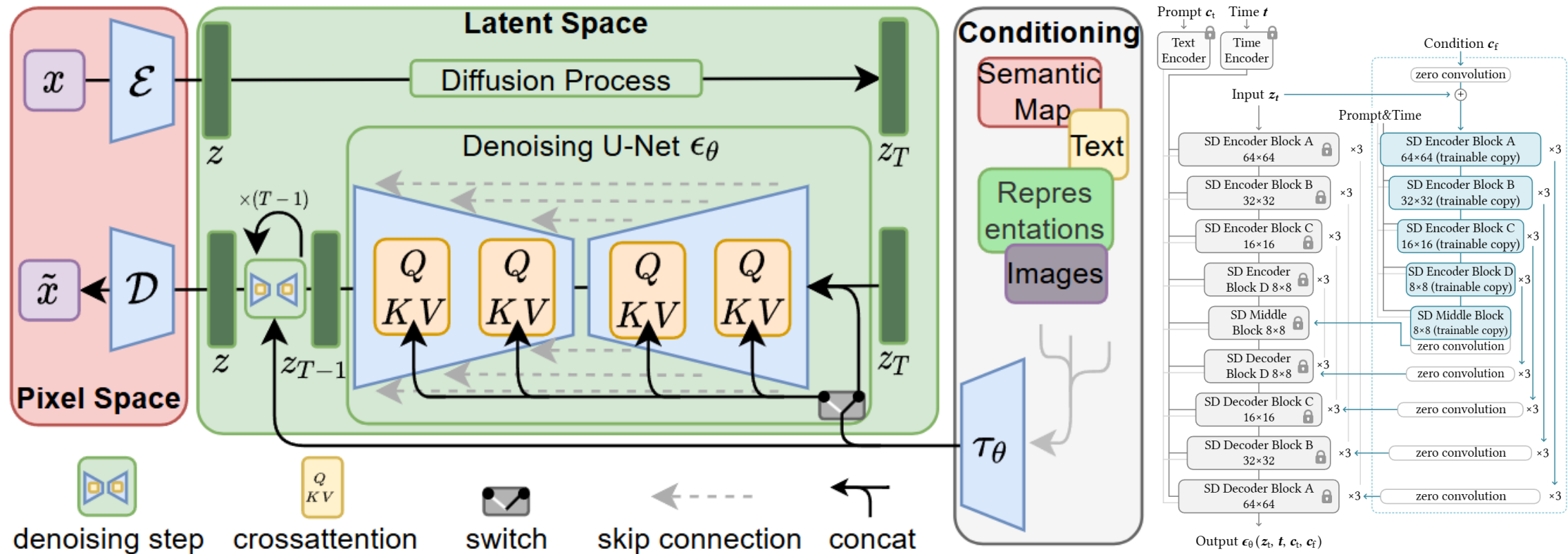
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- What is ControlNet?
- Introduction
- Methods
- Results
- Summary and Discussion

What is ControlNet?



Adding Conditional Control to Text-to-Image Diffusion Models

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What is ControlNet?



Figure 1: Controlling Stable Diffusion with learned conditions. ControlNet allows users to add conditions like Canny edges (top), human pose (bottom), etc., to control the image generation of large pretrained diffusion models. The default results use the prompt "a high-quality, detailed, and professional image". Users can optionally give prompts like the "chef in kitchen".

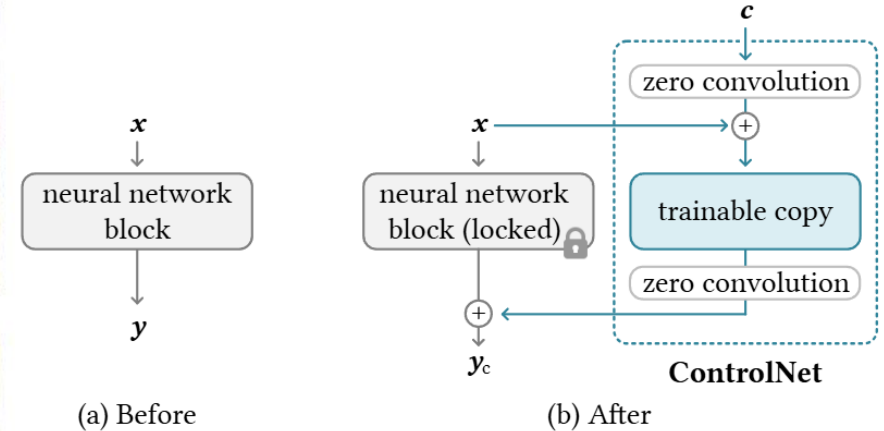


Figure 2: A neural block takes a feature map x as input and outputs another feature map y , as shown in (a). To add a ControlNet to such a block we lock the original block and create a trainable copy and connect them together using zero convolution layers, *i.e.*, 1×1 convolution with both weight and bias initialized to zero. Here c is a conditioning vector that we wish to add to the network, as shown in (b).

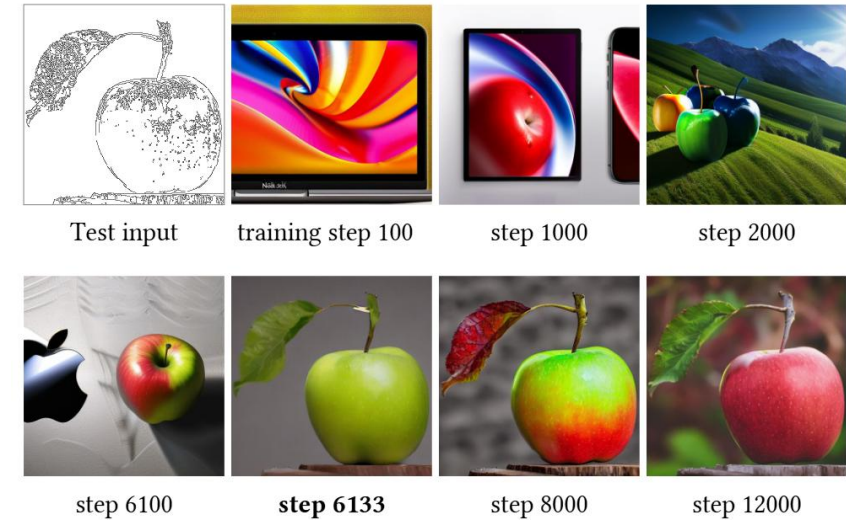


Adding Conditional Control to Text-to-Image Diffusion Models

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What is ControlNet?

1. Locking(freeze)으로 pre-train된 지식을 보존
2. Zero-convolution으로 "점진적이고 안전한" 학습을 보장
3. 다양한 조건(c_f)을 "가볍게 엮을 수 있는 플러그인" 구조로 제작

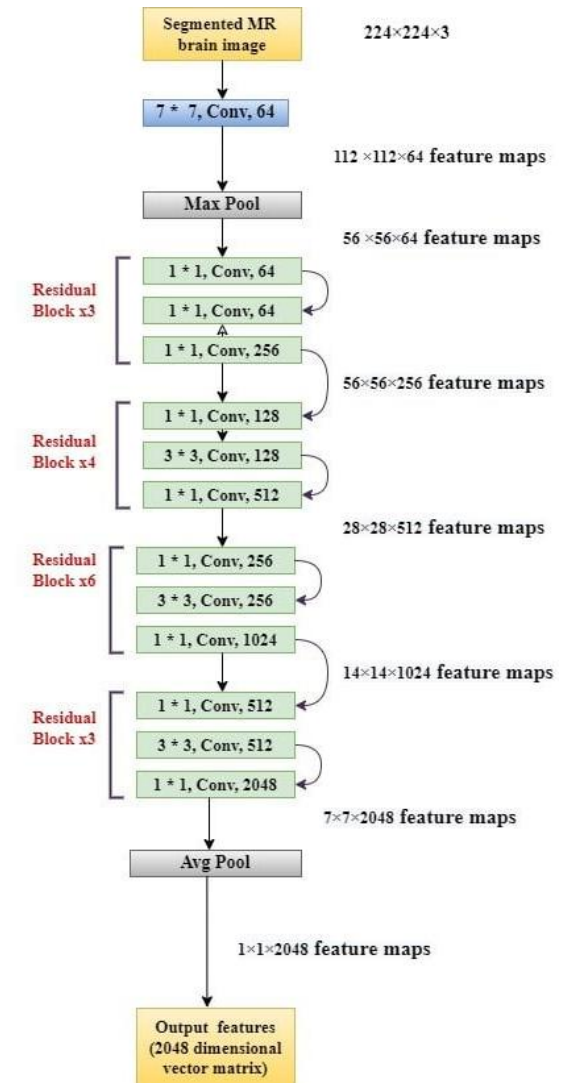
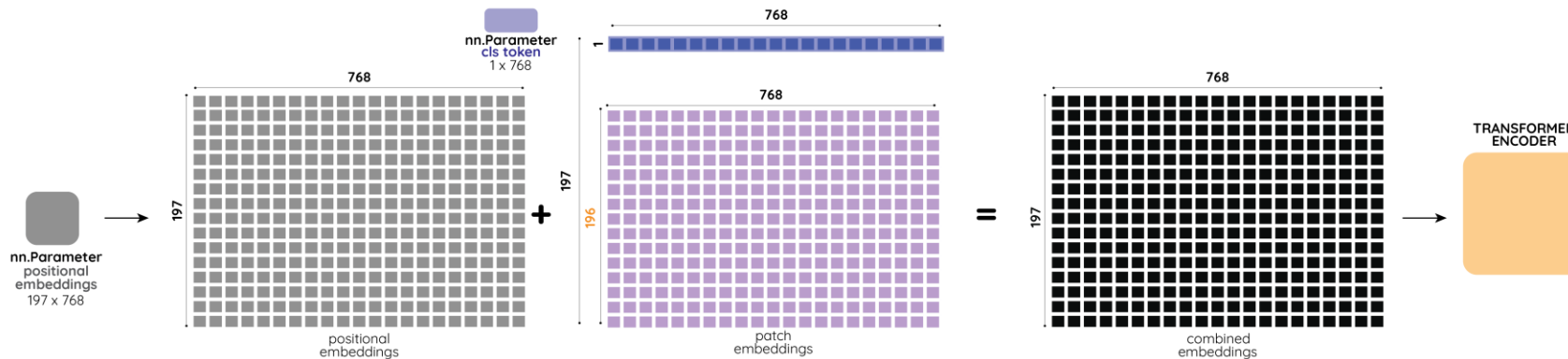


$$\begin{aligned} z_0 &= \text{latent vector}, t = \text{timestep}, c_t = \text{text condition} \\ &+ \\ c_i &= \text{image condition}, c_f = \mathcal{E}(c_i) \\ &= \\ \mathcal{L} &= \mathbb{E}_{z_0, t, c_t, c_f, \epsilon \sim \mathcal{N}(0,1)} \left[\left\| \epsilon - \epsilon_\theta(z_t, t, c_t, c_f) \right\|_2^2 \right] \end{aligned}$$

- Feature space는 neural network 의 decision processes을 이해하는 데 핵심적임.
 - 최근 XAI 연구의 기반이 되지만, 그 중요성은 해석가능성에만 그치지 않음.
- Feature space를 분석하면 모델의 **강건성(robustness)**, **불변성(invariance)**, **대칭성(symmetry)** 같은 근본적인 성질을 연구할 수 있음.
 - RCDM
- Learned representation의 geometry과 topology를 분석함으로써, 모델의 **일반화 능력**, **실패 모드**, 데이터에서 나타나는 **고차원적 패턴에 대한 통찰**을 얻을 수 있음.

- f_θ 의 input과 output의 space가 다른 경우 feature space를 input space와 연결 짓는 것이 어려움.
 - e.g. 이미지 분류 모델: $x \rightarrow y$, where $x \in \mathbb{R}^{H \times W \times C}$, $y \in \mathbb{R}^{\# \text{ labels}}$
- **Localization**
 - feature space의 특정 영역/유닛이 입력의 어느 부분에 반응하는지 찾는 문제.
 - GradCAM_(2017, ICCV): feature map의 관심 영역을 입력 크기로 linear upsampling
 - linear upsampling이 실제 mapping을 정확히 반영한다고 보기 어려움.
- **feature inversion**
 - $\mathbb{R}^{\text{feature space}} \rightarrow \mathbb{R}^{H \times W \times C}$
 - INNs_(2020, ECCV): $z = f_\theta(x)$, $x = f_\theta^{-1}(z)$ bijective한 NN 학습하여 **feature inversion**
 - Deterministic mapping $\rightarrow$ 이 문제가 본질적으로 ill-posed임에도 불확실성을 반영하지 못함.
 - RCDM_(2022, TMLR): Ablated Diffusion Model로 (pooled) **feature inversion**
 - **Pooled feature map** 기반 $\rightarrow$ feature map의 fine-grained spatial detail를 버림.
 - 최신 생성모델을 활용하지 않음.

- ResNet50
 - $\mathbb{R}^{7 \times 7 \times 2048} \rightarrow_{pool} \mathbb{R}^{2048} \rightarrow \mathbb{R}^{224 \times 224 \times 3}$
- ViT-B/16
 - $\mathbb{R}^{(1+196) \times 768} \rightarrow_{[CLS] \text{ or } pool} \mathbb{R}^{768} \rightarrow \mathbb{R}^{224 \times 224 \times 3}$
- Average Pooling
 - 각 patch에 대하여 평균내는 것이기 때문에 spatial information이 손상됨.



- **spatially resolved feature map**을 재구성하지 못하는 것 자체가 **faithfulness 결여**임.
 - XAI나 robustness 분석 같은 downstream application에 치명적임.
 - intermediate layer의 representation을 연구하는 목적에도 맞지 않음.
- **Contribution**
 - ControlNet 스타일 조건부 diffusion model로 spatial feature space → input space로 high-fidelity mapping이 가능함을 보이며 정량적 검증, 강건성/일반화 분석을 함.
 - input space에서 concept steering 시각화, feature space의 composite 특성에 대한 통찰을 제공함.

- Approach

- FeatInv: $\mathbb{R}^{\text{feature space}} \rightarrow \mathbb{R}^{H \times W \times C}$
 - 목표는 **원본 이미지 x 를 픽셀 단위로 정확히 reconstruction하는 게 아님.**
 - 생성된 x 를 다시 모델에 통과시켰을 때, **feature representation이 x 와 align되는 high-fidelity 이미지를 만드는 것이 목표임.**

$z_0 = \text{latent vector}, t = \text{timestep}, f_\phi \in \{\text{ResNet, ConvNext, ViT}\}$

+

$c_t = \text{"a high-quality, detailed, and professional image"}$

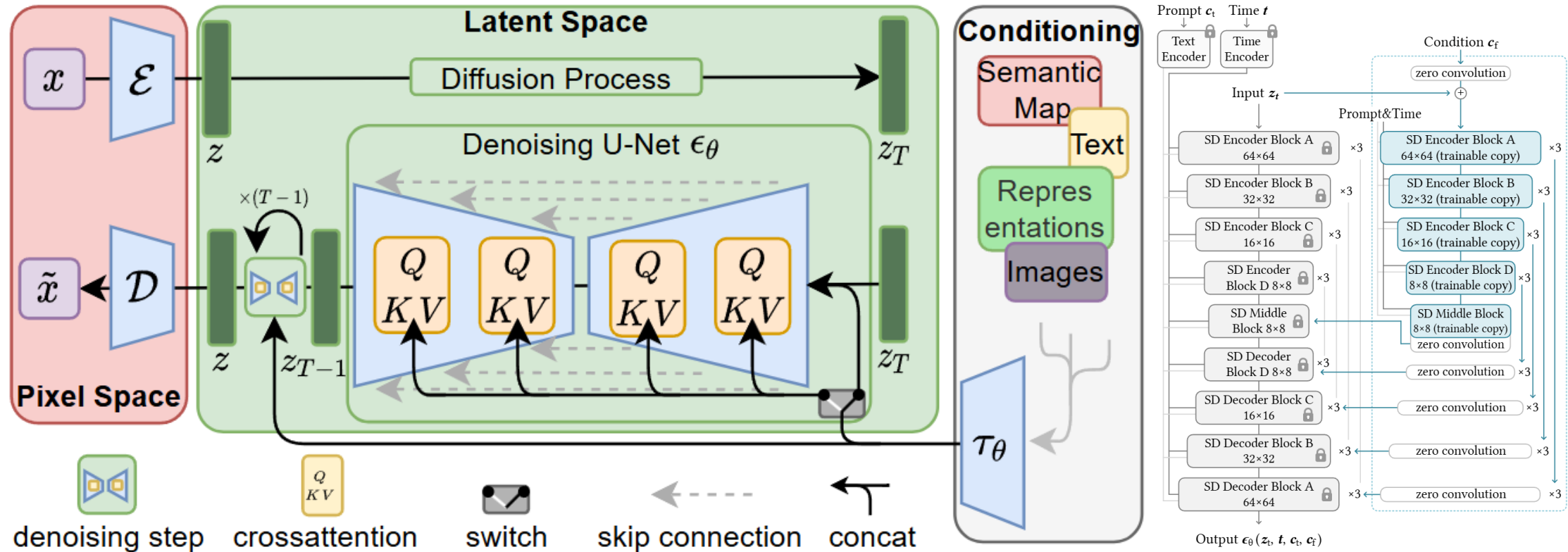
+

$$c_f = f_\phi(x), c_{enc} = \mathcal{E}(c_f)$$

=

$$\mathcal{L} = \mathbb{E}_{z_0, t, c_t, c_{enc}, \epsilon \sim \mathcal{N}(0,1)} [\|\epsilon - \epsilon_\theta(z_t, t, c_t, c_{enc})\|_2^2]$$

- Architecture and training procedure



- Conditional input encoder

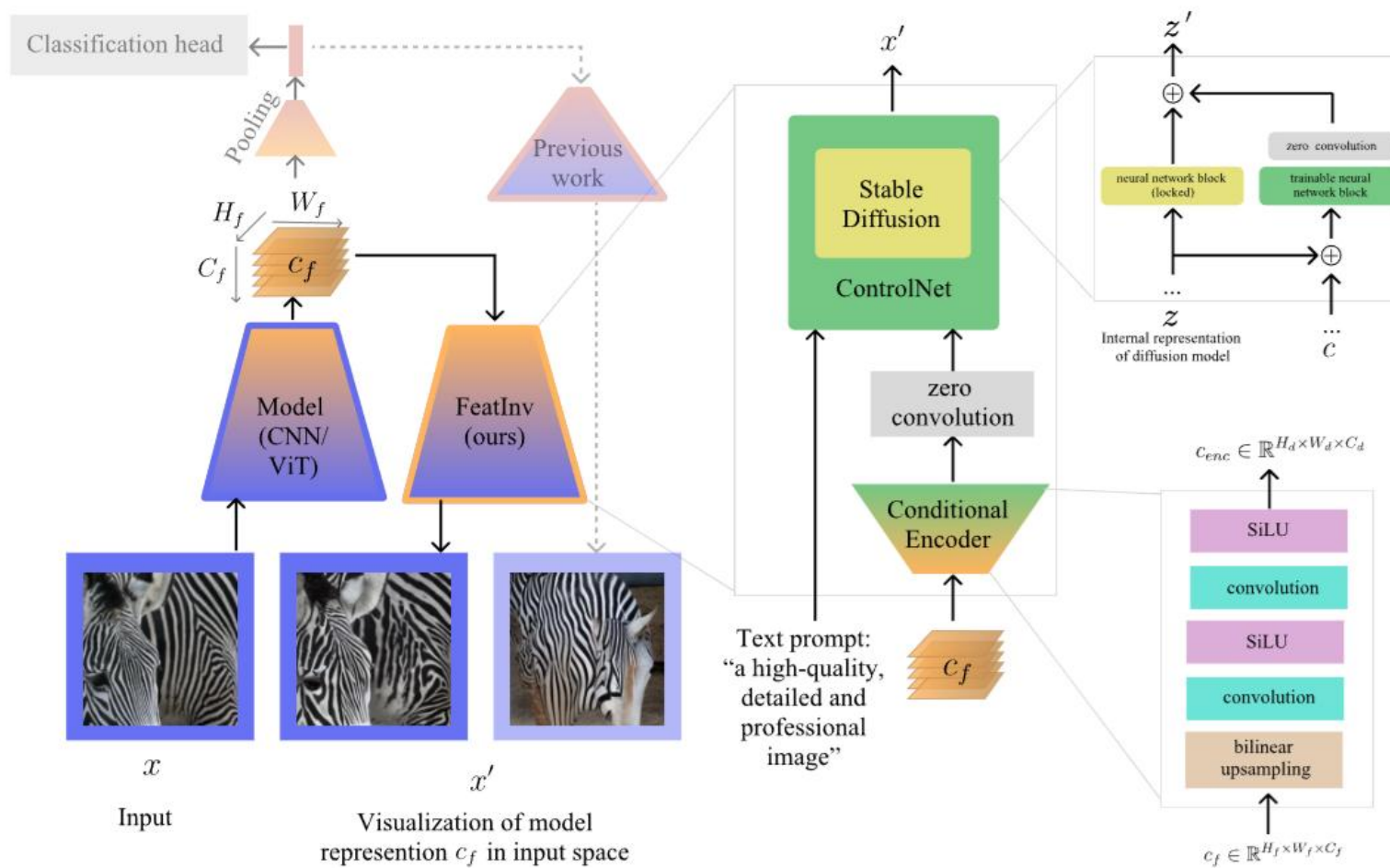
- feature map의 shape (H_f, W_f, C_f)와 diffusion model 내부 표현의 shape (H_d, W_d, C_d)가 다름.
 - ResNet50 : $H_f = W_f = 7, C_f = 2048$
 - MiniSD : $H_d = W_d = 32, C_d = 320$
 - Bilinear upsampling: $7 \times 7 \rightarrow 32 \times 32$ 로 해상도를 맞춤.
 - Shallow CNN: 채널 수 변환($2048 \rightarrow 320$)을 포함해, 모델의 representation space에서 diffusion model의 representation space로 매핑을 학습함.

- Pooled vs. unpooled

- pooled representation은 spatial 정보가 없는 하나의 C_d 차원 벡터임.
- 이를 같은 파이프라인에 넣기 위해, 이 벡터를 $H_d \times W_d$ 만큼 복사해서 강제로 (H_d, W_d, C_d) shape을 만듦.

- Training

- Objective: DDPM (Ho et al. 2020)
- 학습 세팅: ImageNet, batch size 8, lr $1e-5$, AdamW
- 3 epoch, NVIDIA L40 GPU 2장으로 약 45~60시간

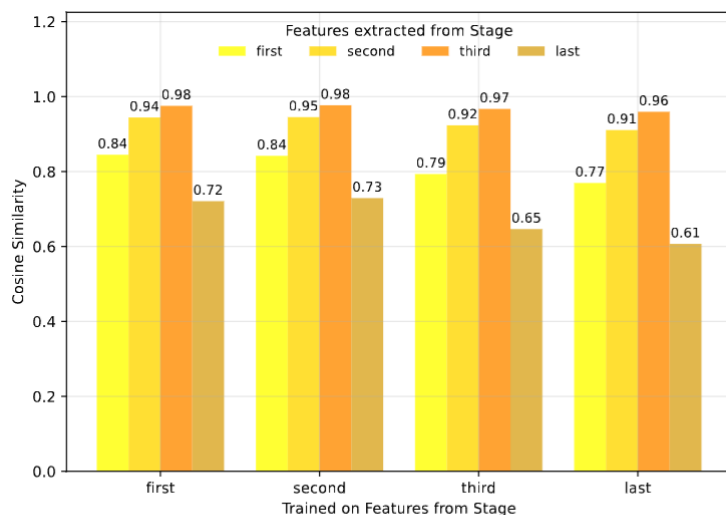


- 실험 세팅
 - **ResNet50, ConvNeXt, SwinV2** (모두 ImageNet-1k에서 pretrained/finetuned)
 - 마지막 hidden layer의 representation을 조건으로 사용함.
 - ImageNet 클래스마다 10개 validation 샘플로 재구성
 - sampling step 50, 모델당 약 12시간 (L40 GPU 1장)
 - feature map 하나당 샘플 1개만 생성
- 평가 기준
 - **Reconstruction quality :**
 - cosine similarity: 재구성된 이미지가 원본과 얼마나 가까이 있는지?
 - top-k match: 재구성된 이미지를 분류기에 넣었을 때 원본과 같은 예측을 내는가?
 - **Sample quality :**
 - 생성된 이미지가 "고충실도 자연 이미지" 집합에 속하는가 → **FID score**로 측정

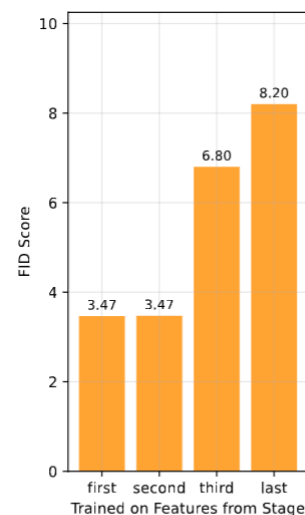
Results - Quantitative and qualitative comparison

	Model	cosine-sim	top5(1) match	FID
unpooled	ResNet50	0.46	91% (70%)	11.49
	ConvNeXt	0.61	94% (77%)	8.20
	SwinV2	0.53	95% (80%)	12.69
pooled	ResNet50	0.12	48% (23%)	31.64
	ConvNeXt	0.19	44% (20%)	31.67
	SwinV2	0.16	47% (22%)	49.04

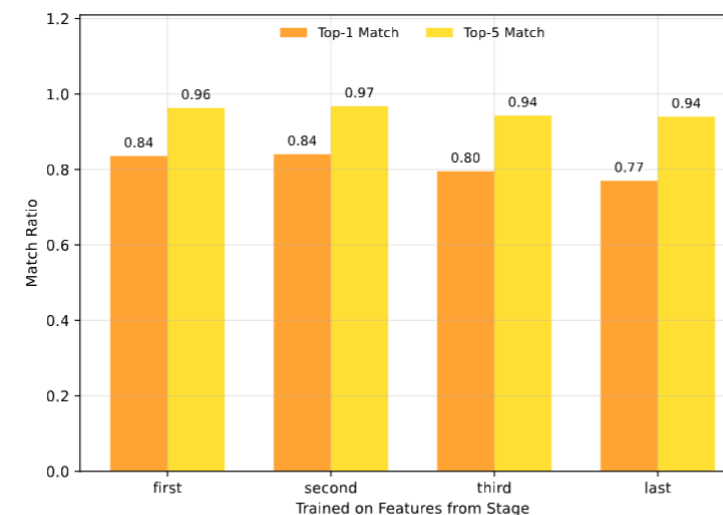
Class-condition's FID ≈ 29



(a) cosine-similarity

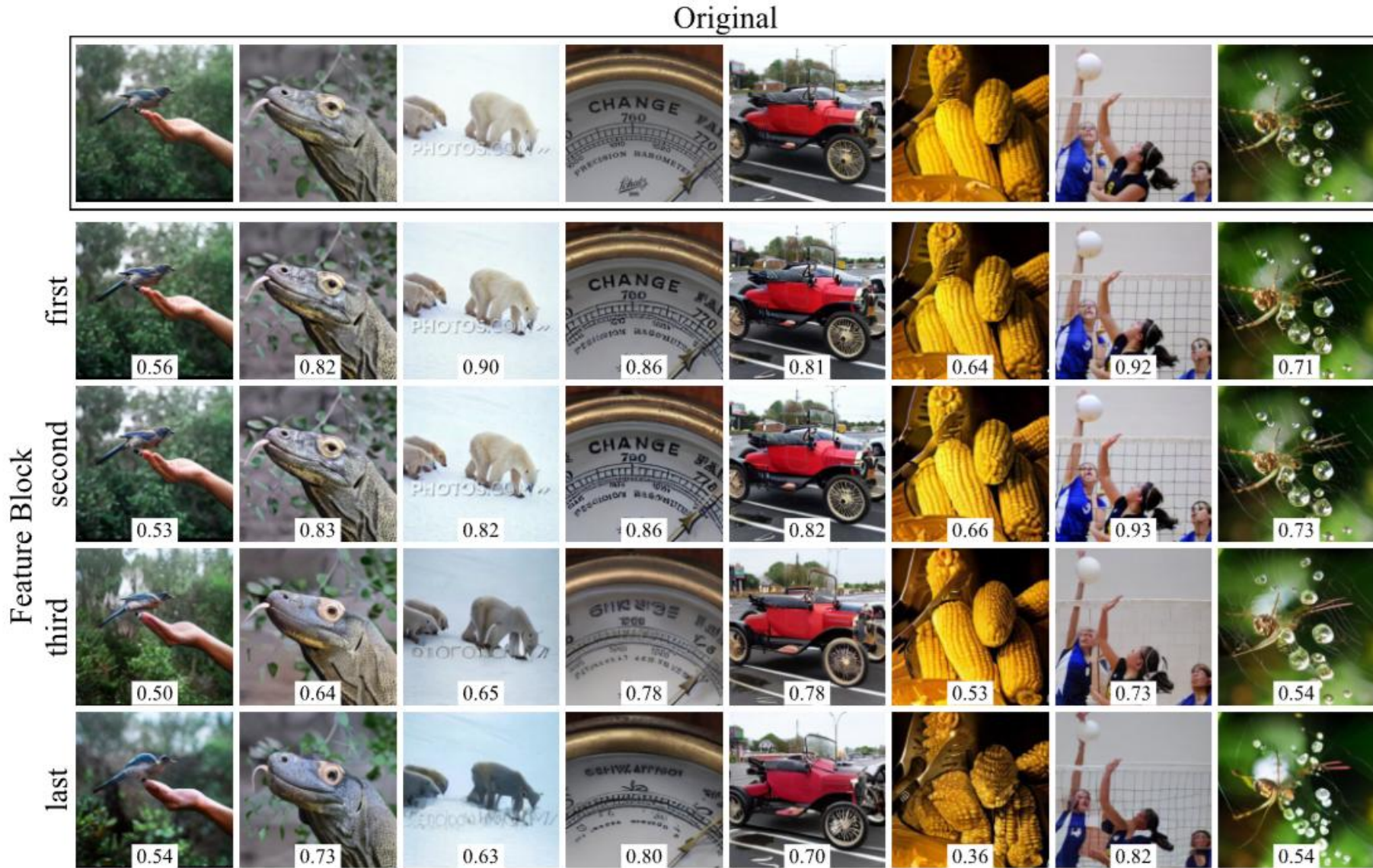


(b) FID score

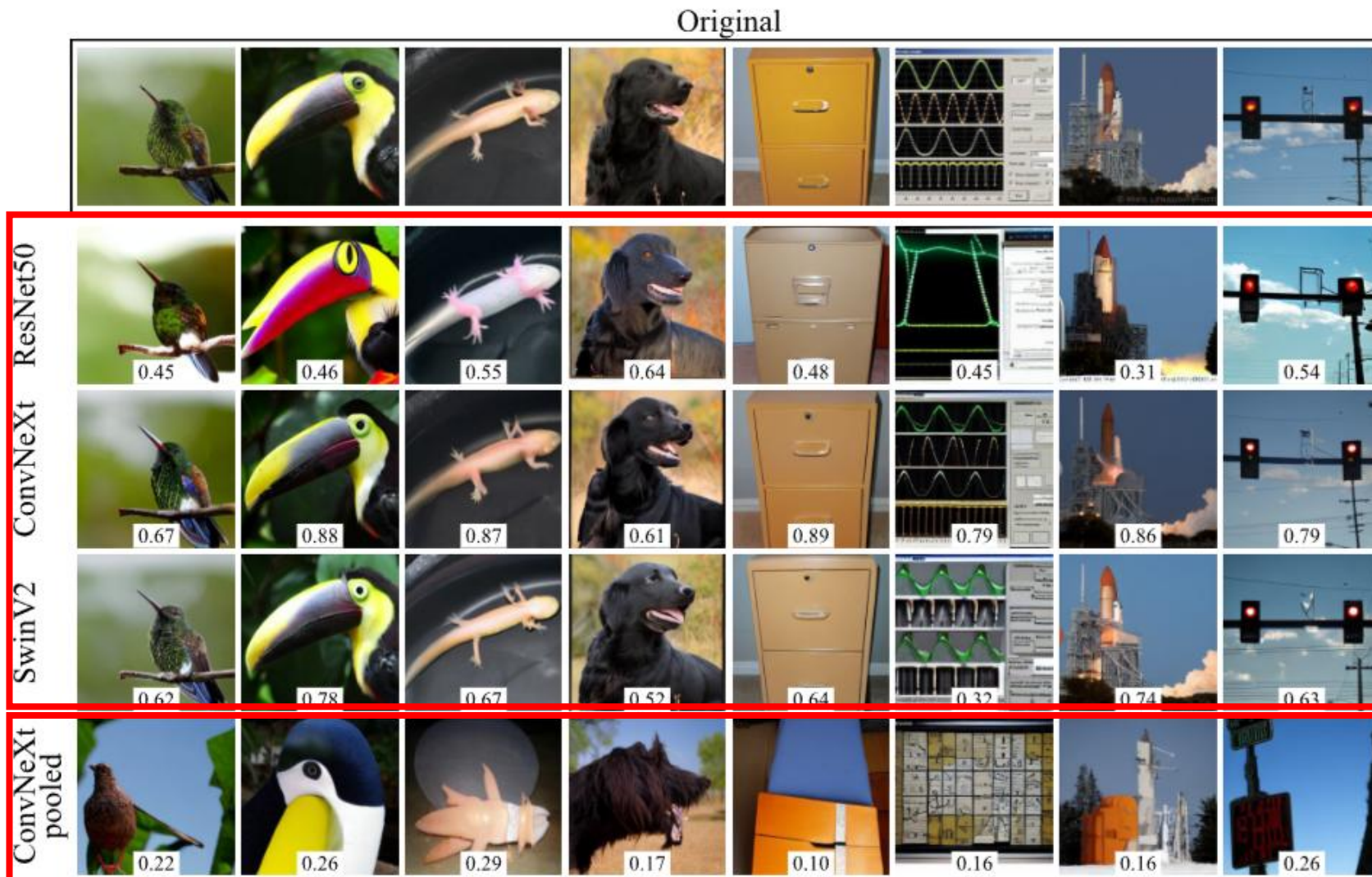


(c) top5(1) matching

Results - Quantitative and qualitative comparison



Results - Quantitative and qualitative comparison



LDM 특성상 high-level semantic feature maps에서 loss를 줬는데도

최신 백본은 색감, 세부 디테일까지 원본과 매우 유사함!

Pooled feature는 Spatial information을 잃어버리므로 구조적인 연구는 신뢰할 수 없다!

- Cross-model evaluation & Evaluation with OOD samples (BAID)
 - cosine-sim = 0.48, FID = 12.98

conditioned on	evaluated by		
	ResNet50	ConvNeXt	SwinV2
ResNet50	91% / 70%	90% / 68%	91% / 71%
ConvNeXt	92% / 73%	94% / 77%	96% / 81%
SwinV2	94% / 77%	94% / 78%	95% / 80%

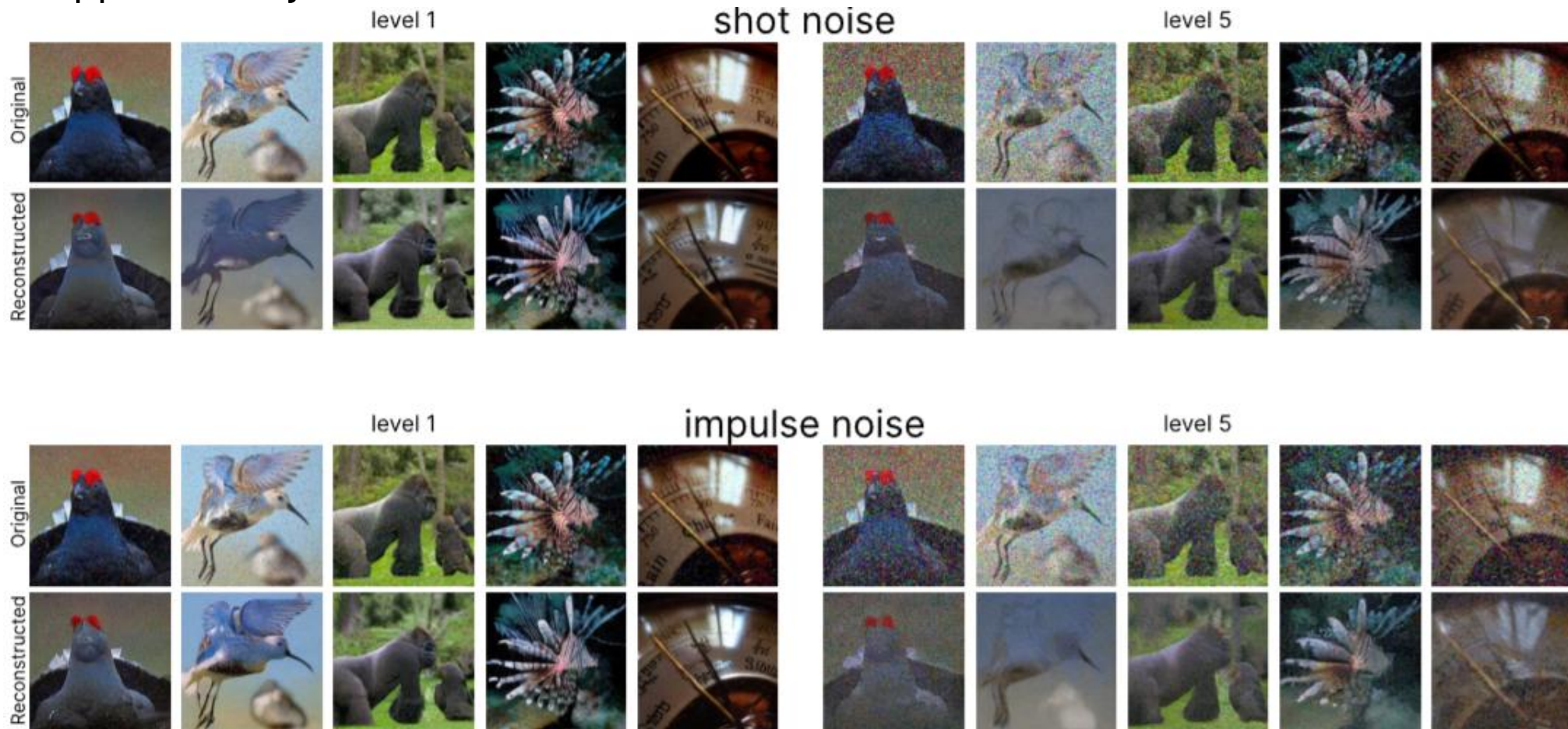
Original



Reconstructed



- Evaluation with adversarial images & Evaluation with corrupted images
 - Supplementary Material C



Feature #34M/31164353 Golden Gate Bridge feature example

The feature activates strongly on English descriptions and associated concepts

in the Presidio at the end (that's the huge park right next to the Golden Gate bridge), perfect. But not all people

repainted, roughly, every dozen years." "while across the country in san francisco, the golden gate bridge was

it is a suspension bridge and has similar coloring, it is often compared to the Golden Gate Bridge in San Francisco, US

They also activate in multiple other languages on the same concepts

ゴールデン・ゲート・ブリッジ、金門橋は、アメリカ西海岸のサンフランシスコ湾と太平洋が接続するゴールデンゲート海

골든게이트 교 또는 금문교는 미국 캘리포니아주 골든게이트 해협에 위치한 현수교이다. 골든게이트 교는 캘리포니아주 샌프란시

мост золотые ворота — висячий мост через пролив золотые ворота. он соединяет город сан-фран

And on relevant images as well



- FeatInv로 feature space $\rightarrow$ input space 역매핑이 가능하니, hidden activation을 조작한 효과를 output(레이블)이 아니라 input representation(이미지) 상에서 관찰할 수 있게 됨.
 - MCD(Multi-dimensional Concept Discovery, Vielhaben et al., 2023)를 사용하여 concept 을 feature space 안의 선형 부분공간(linear subspace)으로 정의함.

Algorithm 1: *FeatInv-Viz*: Visualization of concept steering in input space

Input: Model m , concept decomposition $\phi = \sum_i \phi_i$, concept with id c

Output: Visualization of concept c in input space

Notation: $x \in \mathbb{R}^{3 \times H \times W}$ where $x^{(j)}$ refers to color channels with $j \in \{R, G, B\}$

$\phi' \leftarrow \sum_{i \neq c} \phi_i + 0.25 \cdot \phi_c$; // Attenuated feature map

for $i = 1$ **to** n **do**

$s_i \leftarrow \text{RandomSeed}()$

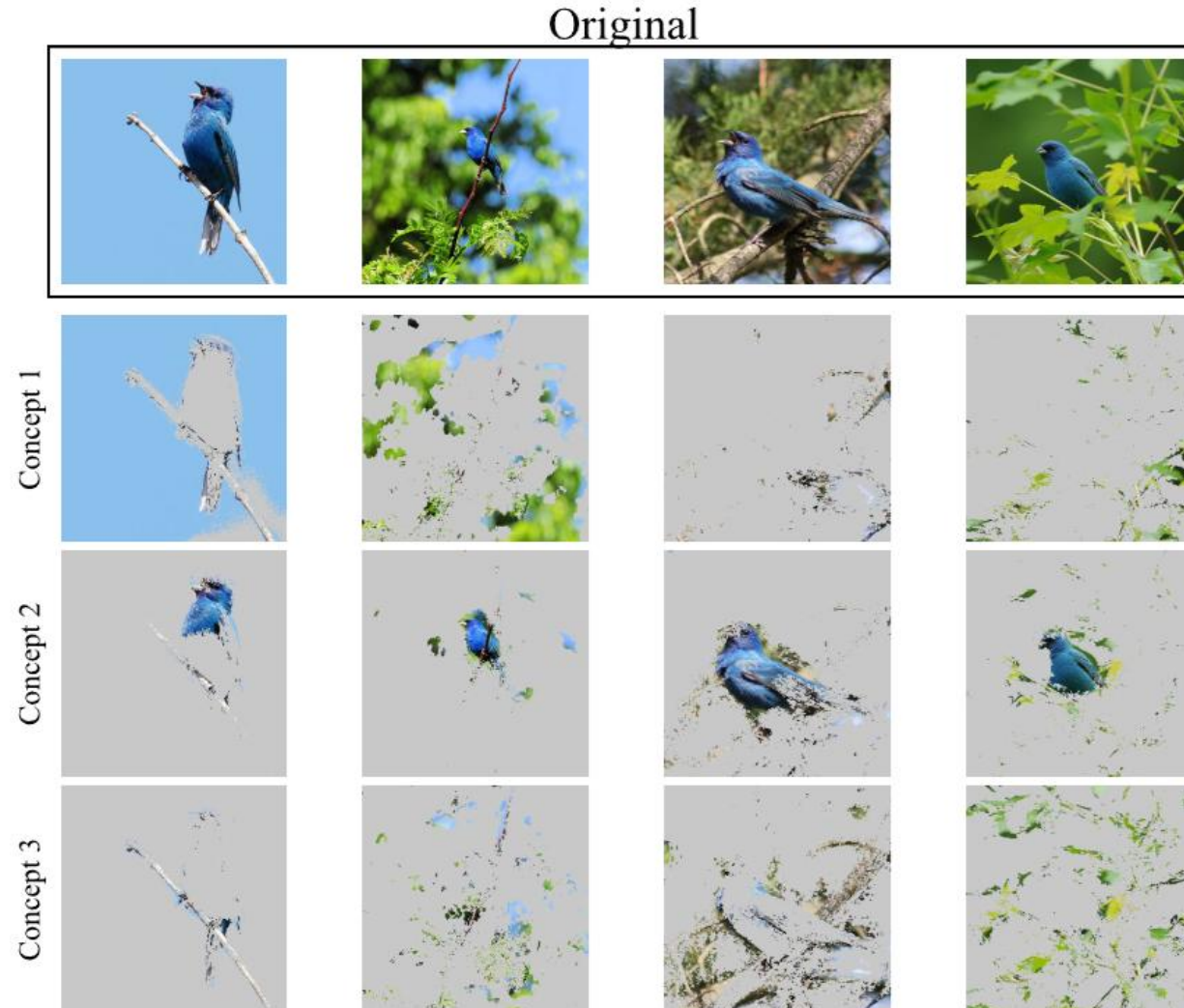
$x_i \leftarrow \text{FeatInv}(\phi, \text{seed} = s_i)$; // Original reconstruction

$x'_i \leftarrow \text{FeatInv}(\phi', \text{seed} = s_i)$; // Attenuated reconstruction

$\Delta_i \leftarrow \sqrt{\sum_{j \in \{R, G, B\}} (x_i^{(j)} - x_i'^{(j)})^2}$; // Euclidean distance

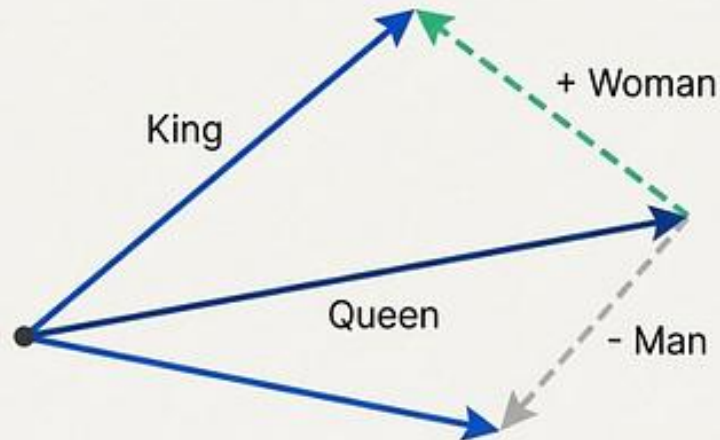
return $\text{median}\{\Delta_i\}_{i=1}^n$; // Median along sample axis

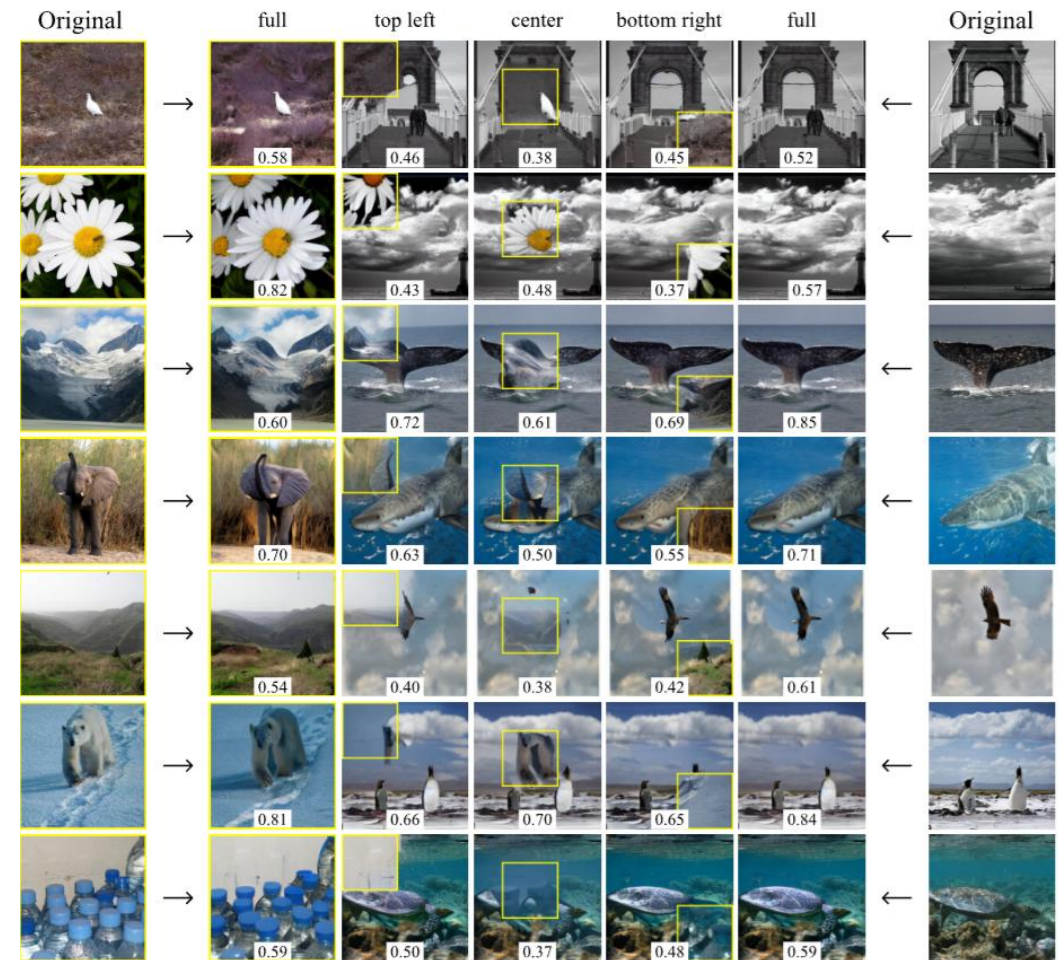
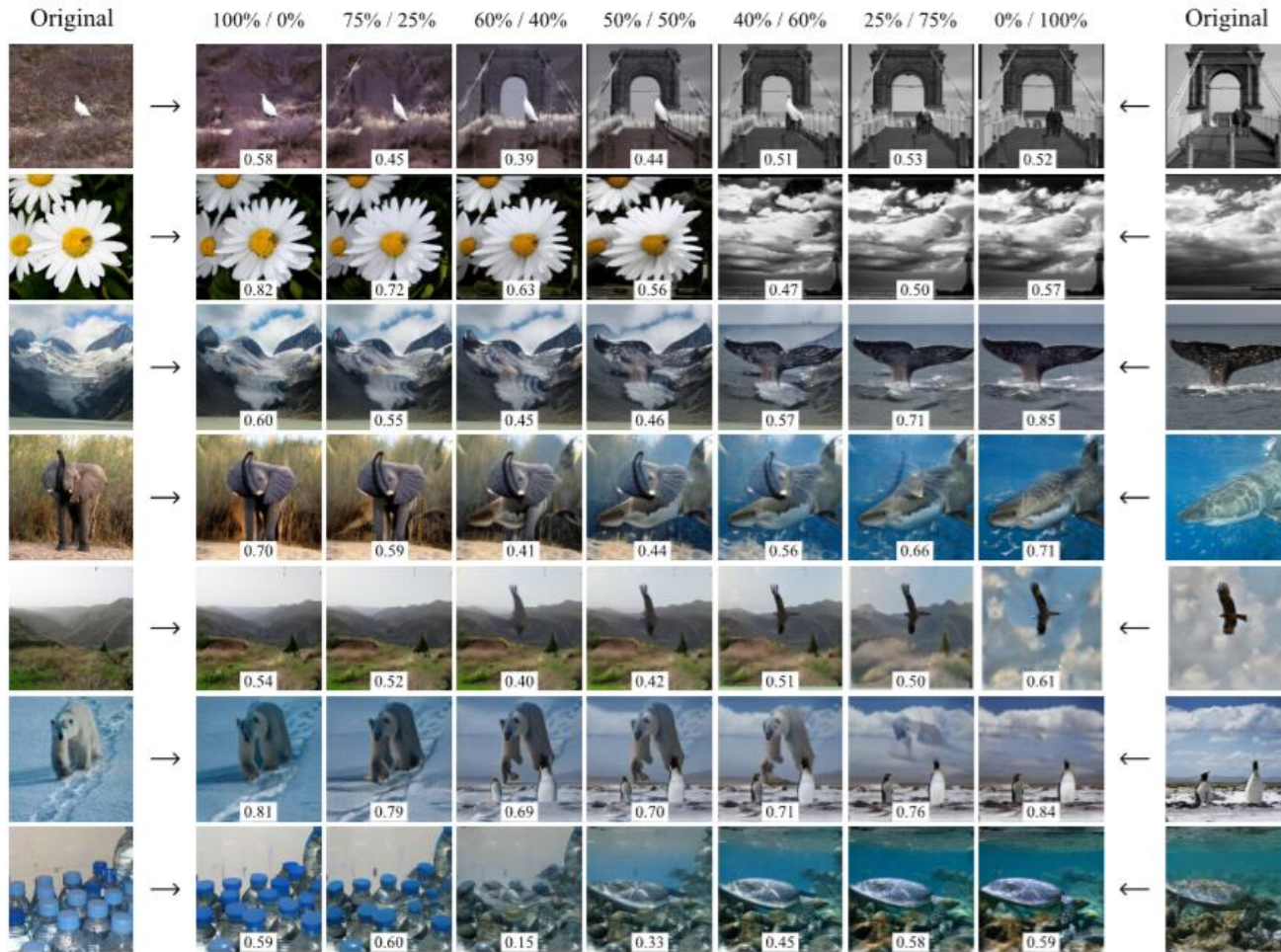
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Capturing Abstract Relationships

The geometry encodes more than just similarity; it captures relationships. Concepts like gender, plurality, and tense exist as consistent directions within the space.





- natural images에서만 검증됨.
 - Medical imaging 등 다양한 도메인으로 확장 필요함.
- pretrained diffusion model 필요함.
 - 다양한 도메인으로 확장성 저해함.
- 모든 모델과 레이어를 학습하기 위해 새로 학습해야 함.
 - Fine-tuning으로 해결 가능해보임.
- 마지막 두 실험이 feature space에서 수정을 통해 얻은 결과임.
 - 신뢰할 수 있는 결과를 얻으려면, 이런 "모델 범위를 벗어난 feature map"을 탐지하는 메커니즘을 도입하는 게 앞으로 중요한 과제임.
- **Grad-CAM의 linear upsampling을 비판했으나 본인도 Bilinear upsampling을 사용함.**
- **Text condition과 기여도 분리가 어려움.**

- 모델의 feature map 구조를 이해하기 위해, conditional diffusion model을 이용하여 **feature space → input space**로의 학습된 확률적(probabilistic) 매핑을 제안함.
 - ControlNet 스타일로 이런 모델을 학습시키는 것이 feasible함을 입증함.
 - 다양한 모델 아키텍처 ResNet50, ConvNeXt, SwinV2에 걸쳐 **정확하고 강건한 재구성**을 달성함.
- feature map을 manipulate한 뒤 그 효과를 들여다보는 방향의 응용 2가지를 제시:
 - FeatInv-Viz (concept steering 시각화)
 - feature space의 composite 성질 탐구 (interpolation/spatial composition)
- concept steering 사례처럼, **모델 이해도를 높임으로써 긍정적인 사회적 영향**을 기대함.